

TarsosDSP: A Real-Time Audio Analysis and Processing Framework

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In last month's EFY Plus, we discussed Essentia, a C++ library for audio analysis. In this issue we will discuss a Java based real-time audio analysis and processing framework known as TarsosDSP.

The tool is counted as one among the few frameworks in Java ecosys-

tem that provides both real-time feature extraction as well as synthesis. The framework consists of easily-extendable practical audio-processing algorithms. Owing to the educational goal of music information retrieval (MIR), these algorithms are made as simple and self-contained as possible using a straightforward pipeline.

Whether you are a student trying to understand the concepts of MIR or an experienced researcher working on music analysis, TarsosDSP is the ideal choice for you. The real-time feature extraction and synthesis capabilities make this software an ideal candidate for music education tools and music video games. Try out the latest release of the software version 2.2 that comes with this month's DVD for EFY Plus.

Implementation in Java ecosystem

The framework, written in Java ecosystem, allows clean implementation of audio-processing algorithms. Compared to the ones written in C++, this Java based framework ensures better portability between various platforms.

The automatic memory-management facilities are yet another advantage imple-

menting the framework in Java. The Dalvik Java runtime allows running the algorithms unmodified on Android platform.

With the exception of the standard Java runtime, there are no other external dependencies. For real-time applications, the operating environment is optimised to provide a low-latency audio pipeline.

Simple processing pipeline

Another notable feature of this framework is its extremely simple processing pipeline. Limiting the input to a single-channel input makes the processing chain very straightforward. A multichannel audio input is automatically down-mixed to a single channel before it is taken to the processing pipeline.

Input samples are chopped and arranged into samples in blocks of variable sizes having a defined overlap, with the help of AudioDispatcher. The blocks are then wrapped and encapsulated in an AudioEvent object having a pointer for timing and some auxiliary methods for computing the energy of the blocks.

AudioEvent is passed through a series of AudioProcessor objects by AudioDispatcher. The core of the algorithms such as pitch estimation or onset detection is carried out in AudioProcessor.

TarsosDSP and competing tools

There are a handful of tools available in the market that work on audio signals.

Essentia software is an open source C++ library that allows us to have real-time music analysis. CLAM tool allows us to have real-time mu-



Fig. 1: Pitch Shifting, an application of TarsosDSP as an example

Framework overview

Name: TarsosDSP

Latest release: Version 2.2

Functionality: Real-time audio analysis and synthesis in Java

Licence: GNU general-public licence (GPL) version 3

For more information: github.com/JorenSix/TarsosDSP

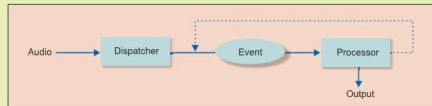


Fig. 2: TarsosDSP processing pipeline, a schematic representation